

Remote work and return to office discussions on Bluesky

Group 25 - Group Report

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Group 25

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1 Introduction

We chose to examine working from home vs returning to the office because it becomes interesting when you consider people’s concerns about well being, costs and job competition. Our aim was to use Bluesky posts to compare sentiment between the two search groups, examine common words and themes, and analyse reply patterns for selected posts.

Our research questions were:

1. How do sentiment categories compare between the return-to-office and remote-work search groups?
2. What vocabulary characterises the English-labelled RTO and WFH search results?
3. Do the English-labelled search results form distinct vocabulary clusters, and how are the two search groups distributed across them?
4. How do accounts interact through replies around our three selected posts?

2 Analysis

We used `atrrr` to collect Bluesky data through the search and thread APIs (Bluesky, n.d.-a, n.d.-b; Gruber et al., 2026). Text processing used `tidytext` (Silge & Robinson, 2016). Clustering used `SnowballC` for stemming (Bouchet-Valat, 2023), TF-IDF weighting (Silge & Robinson, 2017), and R’s `k-means` function (R Core Team, n.d.-a). We used principal components analysis for the cluster display (R Core Team, n.d.-b), `ggplot2` for the text and clustering charts (Wickham, 2016), and `igraph` for network analysis (Csárdi & Nepusz, 2006).

2.1 Data Collection

Research aim: to collect Bluesky posts that let us compare discussion of returning to the office with discussion of remote work.

How the data was accessed

We collected public posts from **Bluesky**, using Bluesky’s open AT Protocol search API through R (`scripts/01_data_collection.R`). We chose Bluesky because posts can be searched programmatically and because workplace policy is actively discussed there. We ran two searches with exact phrases so that the posts are on topic:

- **RTO:** “return to office”.
- **WFH:** “remote work”.

For each search we requested 500 posts, ordered by `latest`. We chose latest because `top` returns only posts that are already popular, which would bias a sentiment comparison towards viral posts. We saved the raw results and labelled each post by the search it came from, so **RTO and WFH describe the search group, not what the author prefers**.

R packages

We used `atrrr` (version 0.2.0) to collect the data, and `dplyr` and `stringr` to clean it. Later sections use `tidytext`, `SnowballC`, `ggplot2` and `igraph`.

Variables collected and data volume

The API returned 592 RTO and 586 WFH posts. For each post it supplies the text, author handle, post address (URI), posting time, like, repost, reply and quote counts, language tags, and the address of the post being replied to.

Cleaning (`scripts/02_data_cleaning.R`) combined the two searches and removed records that cannot be analysed: posts with missing text, posts of 10 characters or fewer (too little language to score), and repeated posts identified by their URI. This left 1,137 posts. The text analysis, clustering and sentiment test use a further subset, the 727 English-labelled posts described below.

Limitations of the data

- **Keyword sampling.** A post is collected only if it contains the exact phrase. Posts using “WFH”, “hybrid work” or “RTO” alone are missed, so the groups are defined by wording and not by every post about the topic.
- **Platform and language.** Bluesky users are not a random sample of workers. Our posts are mostly English and strongly shaped by United States news, particularly government workplace policy, so the results do not describe Australian workers or the public in general.
- **Short, uneven time windows.** We requested the latest 500 results for each phrase, so the groups cover different periods (see below). This affects direct comparisons between them.
- **Not every post is an opinion.** Some posts are job adverts, news links or unrelated uses of the phrase. The later sections show that these shape the apparent differences between groups.
- **Frozen data.** Searching again would return different posts, so the keyword data was collected once and every analysis uses the saved files in `data/`. The three conversation threads used for the network in Section 2.5 were collected separately, and are also saved.

The saved raw files contain 592 RTO and 586 WFH records. The cleaning script combines RTO first, so a URI appearing in both searches retains the RTO label. The resulting saved dataset contains 1,137 posts (577 RTO, 560 WFH).

Saved RTO post timestamps span 25 August to 29 September 2026; WFH timestamps span 15 to 29 September 2026 (UTC, ignoring missing timestamps). These are observed post-date ranges, not guaranteed collection windows. The different time coverage and latest-result selection limit direct population comparisons. This analysis reads the frozen files and does not collect fresh data.

For the text and clustering sections, we retain only posts with at least one language tag and with **every tag equal to en or an English regional variant** (case-insensitive). This leaves 751 English-labelled posts. We exclude 345 records without language tags and 41 with non-English or mixed tags. This is a metadata-based rule, not independent language verification; some excluded unlabelled posts may be English, and tagged posts may be mislabelled.

URLs, including bare domains and query strings, and account mentions are removed. Text is tokenised into lowercase words; English stop words, tokens containing non-ASCII letters or nonletters, and one-letter tokens are removed. Twenty-four English-labelled posts have no retained tokens, leaving **727 usable posts: 371 RTO and 356 WFH**. Full word forms are retained for readable descriptive charts. The per-post audit is saved in `outputs/text/inclusion_audit.csv`.

Table 1: Text-analysis inclusion decisions by search group

group	decision	posts
RTO	Included	371
RTO	Missing language tag	167
RTO	No retained English word tokens	17

group	decision	posts
RTO	Non-English or mixed language tags	22
WFH	Included	356
WFH	Missing language tag	178
WFH	No retained English word tokens	7
WFH	Non-English or mixed language tags	19

The earlier 20-post review identified political-office references, a Swedish post and link-only material. It was a diagnostic sample, not a complete relevance screen. No topic-based exclusions or text-based deduplication are applied here: distinct URIs can repeat the same message. Results therefore describe **English-labelled search results**, not a verified corpus of workplace-only discussion. The network section uses its own separately collected thread sample; these text-analysis filters do not alter it.

2.2 Hypothesis Testing

Research question: How do sentiment categories compare between the return-to-office and remote-work search groups?

To answer this with a formal test, we ask whether sentiment (negative, neutral or positive) is associated with the search group a post came from. As in the other sections, RTO and WFH describe the search, not the author’s preference, and we use the same English-labelled posts.

- **H0:** Search group and sentiment are independent. Knowing the group tells us nothing about the sentiment.
- **H1:** Search group and sentiment are not independent. There is an association.

Method

We used the same 727 English-labelled posts as the text analysis and clustering, so the sections describe the same data. We scored sentiment with the Bing lexicon of Hu and Liu (2004), supplied by `tidytext`, which labels words as positive or negative. We split each post into words and counted its positive and negative lexicon words. A post was classed as *Positive* if it had more positive than negative words, *Negative* if it had more negative than positive words, and *Neutral* if the counts were equal.

Words removed before scoring. Our search phrases are part of the data. The Bing lexicon counts “work” as positive, and many WFH posts contain “remote work”, so scoring it could make WFH posts look positive before any opinion is expressed. We therefore removed the eight query words used in the text analysis (return, office, remote, work, home, rto, wfh and remotework) from the dictionary before scoring. We also removed “trump”, which Bing counts as a positive verb but which here is almost always a name. The table shows how often the dictionary would have scored these words.

Table 2: Removed words that the Bing lexicon would otherwise have scored

Search group	Word	Bing label	Occurrences
RTO	trump	positive	36
RTO	work	positive	43
WFH	trump	positive	2
WFH	work	positive	387

Posts containing no remaining lexicon words cannot be scored, so they are not in the test: 548 of the 727 posts were scored (75.4%).

Both variables are categorical (search group and sentiment class), so we used a **chi-squared test of independence** at a significance level of 0.05. The test compares the observed counts with the counts expected if group and sentiment were independent. The expected count in each cell is (row total x column total) / grand total. The test statistic is the sum of $(\text{observed} - \text{expected})^2 / \text{expected}$ over all cells, with $(\text{rows} - 1) \times (\text{columns} - 1) = 2$ degrees of freedom. The test needs expected counts of about 5 or more; the smallest here was 39.3, so the assumption holds. We also ran a simulated p-value, as in the lecture slides.

Results

Table 3: Observed counts of scored English-labelled posts by search group and sentiment

	Negative	Neutral	Positive	Total
RTO	145	32	102	279
WFH	72	48	149	269

Table 4: Row percentages: sentiment within each search group

	Negative	Neutral	Positive
RTO	52.0	11.5	36.6
WFH	26.8	17.8	55.4

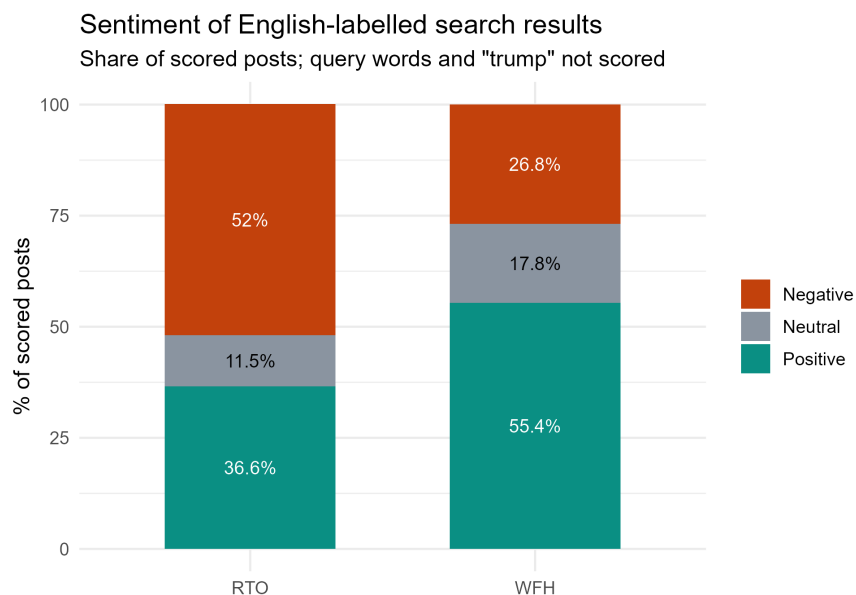


Figure 1: Sentiment of scored posts within each search group.

The chi-squared test gave $X^2 = 36.39$, $df = 2$, $p < .001$. The simulated p-value was $p < .001$. Cramer's V, the effect size, was 0.26.

Among the 279 scored RTO posts, 52.0% were negative and 36.6% positive. Among the 269 scored WFH posts, 26.8% were negative and 55.4% positive.

Table 5: Standardised residuals (values beyond ± 2 show cells that differ clearly from independence)

	Negative	Neutral	Positive
RTO	6.03	-2.11	-4.42
WFH	-6.03	2.11	4.42

Interpretation

Because $p < .001$ is below 0.05, we **reject H0**. There is statistically significant evidence that search group and sentiment are associated: the sentiment mix of the RTO posts differs from that of the WFH posts by more than chance would explain. The standardised residuals show where the difference lies. Cramer’s V of 0.26 indicates a small to moderate association.

This is an **association, not a cause**. The test does not show that remote work makes people happier or that return-to-office policies make them angry. The two labels describe the search each post came from, not the author’s stated preference.

Table 6: Words that drove the sentiment scores

Search group	Sentiment	Most frequent lexicon words (count)
RTO	negative	bribery (10), struggling (10), lost (7), shit (7), corruption (6)
RTO	positive	like (15), boom (12), well (11), good (10), promised (10)
WFH	negative	disabled (20), urgent (10), bad (6), lost (5), shit (5)
WFH	positive	good (32), like (21), enough (17), great (16), recover (13)

The table shows what the scores rest on, and it shows why sentiment here is not the same as opinion. Several of the negative words, such as “bribery” and “corruption” in the RTO posts and “disabled” and “urgent” in the WFH posts, come from repeated news stories and appeals found in the clustering analysis, not from people judging a work arrangement. Words such as “like” and “well” can also be positive in the dictionary without expressing approval.

Robustness checks

Without removing the query words. Scoring the same posts without removing the query words or “trump” gave X-squared = 131.77, $df = 2$, $p < .001$. RTO posts were then 46.1% negative and 44.0% positive, and WFH posts 6.7% negative and 80.4% positive. Comparing this with the main result shows how much the dictionary’s treatment of the search words alone changes the picture.

All cleaned posts. The first analysis used all 1,137 cleaned posts, not only the English-labelled ones, and did not remove any words. That test gave X-squared = 212.08, $df = 2$, $p < .001$. The conclusion is the same as for the main analysis.

Job adverts. Many WFH posts looked like job adverts, which use positive words and say nothing about how people feel about remote work. A keyword rule (for example “hiring”, “apply now”, “careers”, “remotejobs”) flagged 25 WFH posts (7.0%) and 4 RTO posts (1.1%). After removing them, X-squared = 33.25, $df = 2$, $p < .001$. WFH posts were then 54.5% positive and 27.5% negative, and RTO posts 36.7% positive and 52.0% negative. The association remained statistically significant, so job adverts alone do not explain the main result.

Limitations

- **Dictionary sentiment is crude.** The Bing lexicon scores single words. It misses sarcasm and negation (“not good”) and cannot tell opinion from news. Removing the query words fixes one known bias, but other words can still be scored for reasons unrelated to how the author feels.
- **Unscored posts are left out.** 179 English-labelled posts had no remaining lexicon words and were excluded, and they were more common in the RTO group (92 posts) than in the WFH group (87). This could bias the result if those posts differ in tone.
- **Post type differs between groups.** The text analysis and clustering found that RTO posts often discuss policy and political news, while WFH posts include job adverts, appeals and repeated messages. Part of the sentiment difference may reflect the kind of post, not how the author feels. Repeated messages from different accounts are counted as separate posts.
- **Different time coverage.** The English-labelled RTO posts span 25 August to 29 September 2026, but the WFH posts only 15 September to 29 September 2026. News events in the longer RTO window may affect its sentiment.
- **Language tags are only a filter.** The English subset relies on Bluesky language tags, and many posts have none.
- **Independence.** The test assumes each post is independent, but the same accounts can post more than once.
- **Significance is not importance.** With about 548 posts a modest difference can be significant, so we report the effect size and percentages as well as the p-value.

2.3 Text/Content Analysis and Visualisation

Research question: What vocabulary characterises the English-labelled RTO and WFH search results?

We use two complementary measures. Raw frequency counts every occurrence of a word; the post-level percentage counts a word only once per post and divides by the number of usable posts in that search group. Thus a word repeated several times in one post increases raw frequency but not post-level prevalence. These are descriptive comparisons, not hypothesis tests.

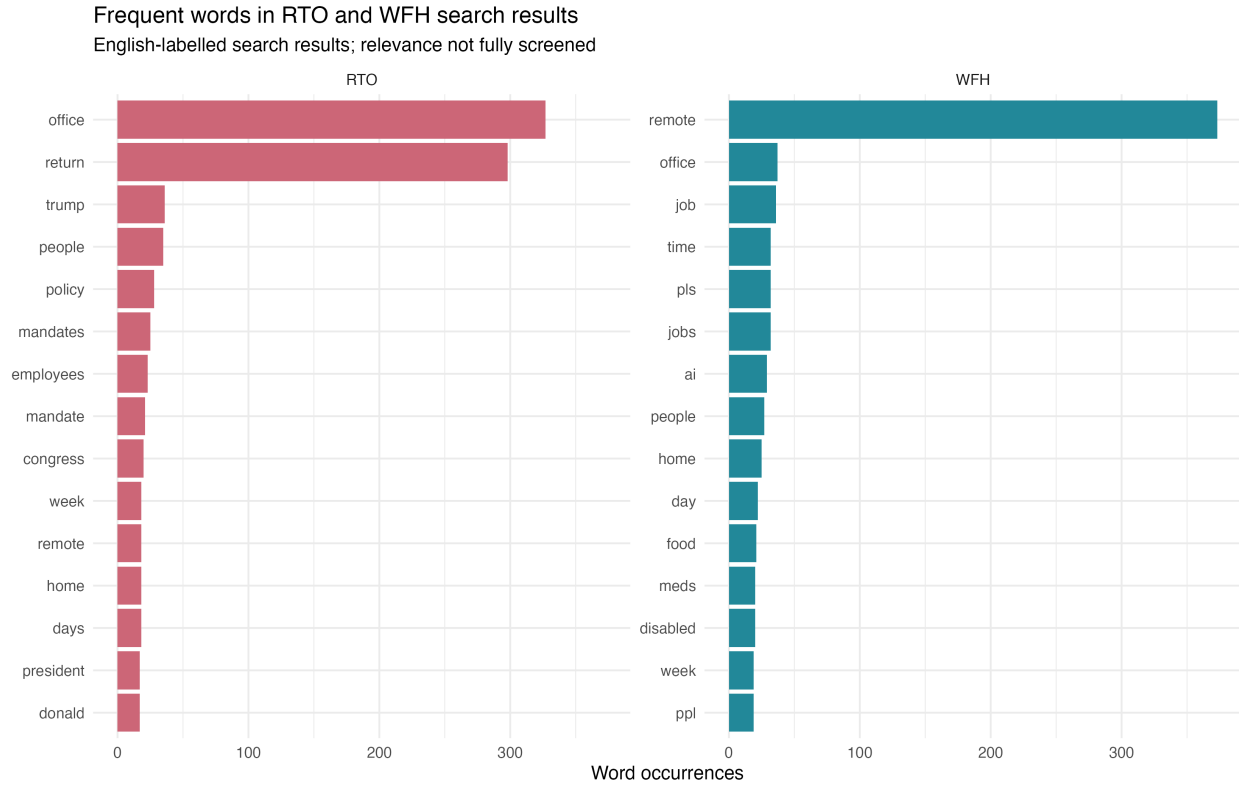


Figure 2: Top 15 words per search group by raw occurrences. Panels share a common count scale.

The frequent-word figure is dominated by terms connected to the queries, such as “return”, “office” and “remote”. This is partly a consequence of how the posts were retrieved and should not be presented as an independently discovered topic.

For the second figure only, we exclude **return**, **office**, **remote**, **work**, **home**, **rto**, **wfh** and **remotework**. Both panels use the same percentage scale; their denominators are 371 RTO posts and 356 WFH posts. Each panel shows its own top 15 terms, so an absent word is not necessarily absent from that group.

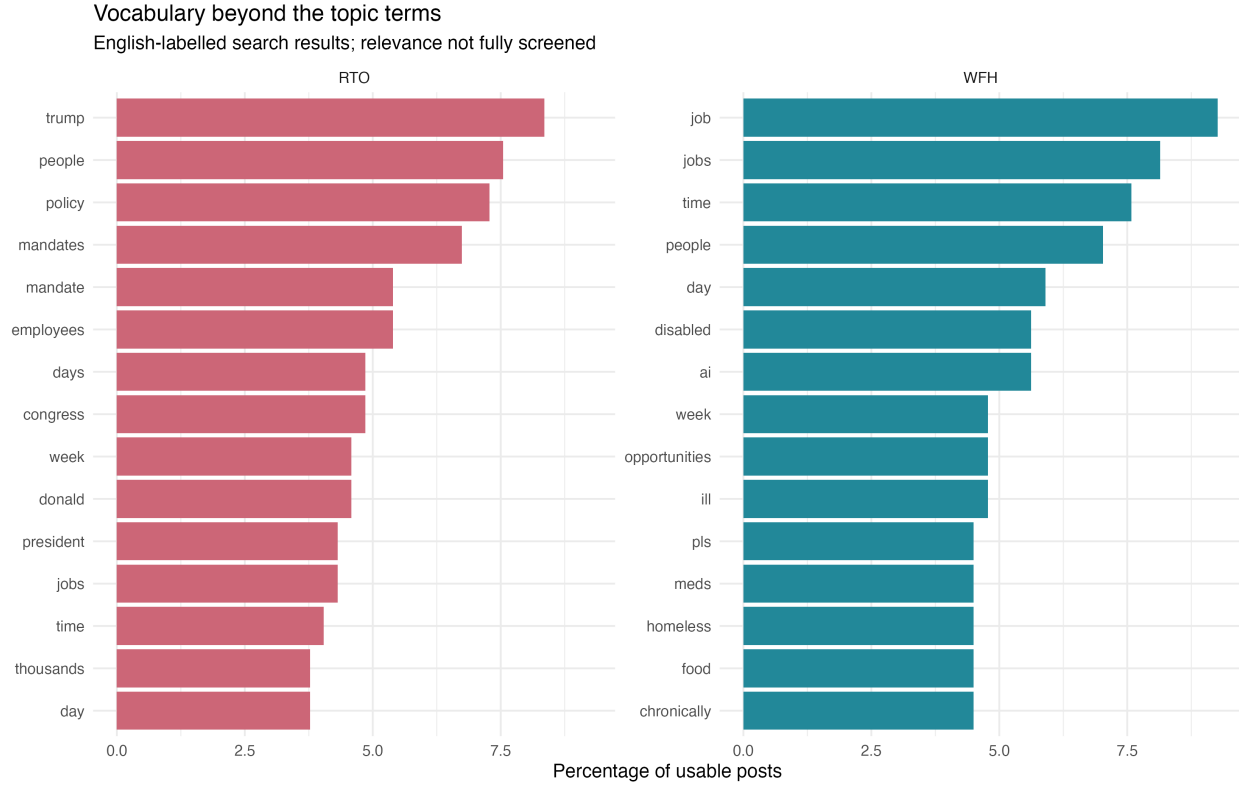


Figure 3: Percentage of usable posts mentioning each word, excluding topic-defining words. Word forms such as job/jobs and mandate/mandates are counted separately.

In RTO results, “policy” appears in 27 posts (7.3%), “mandates” in 25 (6.7%) and “employees” in 20 (5.4%). WFH results include “job” in 33 posts (9.3%), “jobs” in 29 (8.1%) and “time” in 27 (7.6%). These patterns suggest policy/employment vocabulary in RTO and job-related vocabulary in WFH. They do not establish support for either arrangement or the sentiment of the posts.

“Trump” occurs in 31 RTO posts (8.4%). Some political terms refer to genuine government workplace policies, while others come from unrelated political-office stories. Removing names alone would not resolve that distinction. Similarly, recruitment and repeated fundraising messages can contribute to WFH word counts. This motivates the clustering analysis: are the apparent themes broad patterns, or are some driven by repeated message families?

2.4 Clustering

Research question: Do the English-labelled search results form distinct vocabulary clusters, and how are the two search groups distributed across them?

Representation and method

Starting with the same 727 usable posts, we remove the eight topic-defining terms used in the second word chart and apply English stemming with `SnowballC` (for example, employee/employees share a stem). We retain stems appearing in at least three posts to reduce one-off names and idiosyncratic vocabulary. This threshold is a modelling choice that can suppress rare themes. Twenty-seven posts become empty after these additional filters; all are recorded in `outputs/clustering/excluded_posts.csv`. The clustering matrix contains **700 posts and 811 stems**.

Term frequency is a stem's count divided by the total count of retained stems in that post. Inverse document frequency is $\log(N / \text{df})$, where $N = 700$ and df is the number of retained posts containing that stem. Their product is TF-IDF. We normalise each document vector to unit Euclidean length to reduce the effect of post length.

We run ordinary k-means using squared Euclidean distance on these normalised vectors, following the course's TF-IDF, normalisation and k-means concepts. For two unit document vectors, squared Euclidean distance equals twice cosine dissimilarity. However, k-means centroids are not constrained to unit length, so this is **not spherical k-means or direct optimisation of cosine distance**. No two-dimensional projection is used to fit the clusters, and RTO/WFH labels are excluded from the input features.

Number of clusters

For each k from 1 to 10, we use 30 random starts, at most 100 iterations and the reproducible seed 3020 + k . We examine within-cluster sums of squares and representative text. The elbow curve has **no decisive bend**, so there is no strong evidence for one natural cluster count.

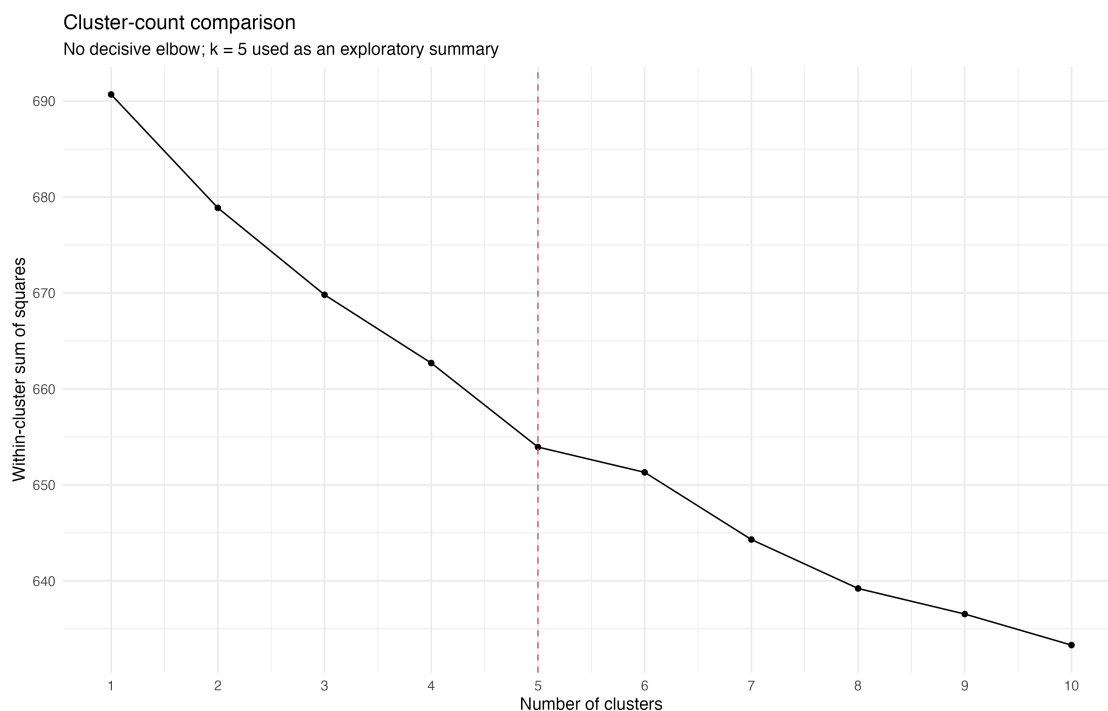


Figure 4: Within-cluster sums of squares for $k = 1$ to 10. The dashed line marks the exploratory five-cluster solution.

We select $k = 5$ as an **interpretable exploratory summary**. In the three-cluster run, 11 job-referral posts and ten auto-manufacturing news posts were combined. Five clusters separate these repeated families and also isolate another political-news group. The improvement from $k = 4$ to 5 is about 8.76 sum-of-squares units, compared with 2.63 from $k = 5$ to 6, but later reductions still occur. This is a pragmatic interpretation choice, not a claim of five well-separated workplace themes. The five-cluster solution accounts for only **5.32%** of total sum of squares between clusters.

Cluster contents and search-group composition

We inspect the highest mean TF-IDF stems and actual posts, including the small clusters’ message families. Cluster numbers are ordered by decreasing size. Labels below are analyst descriptions; the algorithm supplies numbers, not meanings.

Cluster	Posts	RTO	WFH	Interpretation
1	642	327	315	Broad mixed remainder: workplace, policy, employment and other content
2	21	21	0	Repeated political-news wording, including stock-trading and investigation stories
3	16	0	16	Repeated fundraising/assistance appeals mentioning remote work
4	11	0	11	Repeated remote-job referral advertisements
5	10	10	0	Repeated auto-manufacturing news mentioning a president’s return to office

Cluster 1 contains **91.7%** of the clustered posts. Its terms include **peopl**, **mandat**, **time**, **day**, **polici**, **job**, **employe** and **worker**. Nearby example posts discuss commuting costs, unequal access to home working and return-to-office requirements. This large cluster is heterogeneous and should not be treated as a single coherent theme.

Cluster 2 contains political-news language such as **trade**, **congress**, **stock**, **fbi** and **collin**. These describe the content of the posts, not independently verified news claims. Cluster 3 contains **pls**, **food**, **med**, **ppl** and **chronic**, reflecting repeated requests for assistance. Cluster 4 contains **cash**, **luck**, **mercor**, **extra** and **opportun**, reflecting job-referral promotions. Cluster 5 contains **reviv**, **promis**, **wave**, **struggl** and **upend**, reflecting repeated wording from one news story. The political-office families demonstrate query ambiguity rather than workplace preferences.

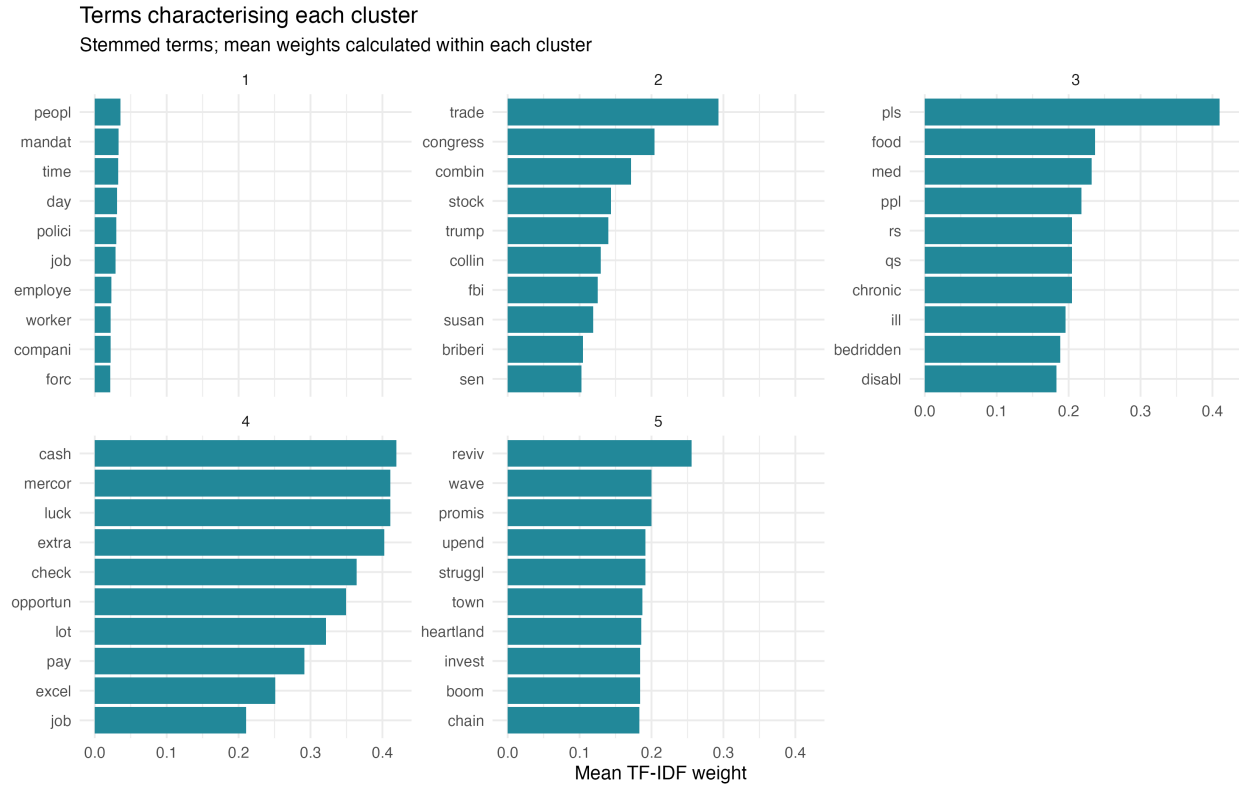


Figure 5: Ten highest mean TF-IDF stems per cluster. Stemming produces shortened forms; weights are averaged within each cluster.

The 11 referral posts have only three distinct cleaned token sequences; the 16 assistance posts have 13. These exact-sequence counts understate near-duplication because small wording changes create different sequences. Distinct URIs were retained, so repetition can drive a cluster. Cluster sizes are post counts, not counts of independent people or opinions.

Visualisation and interpretation limits

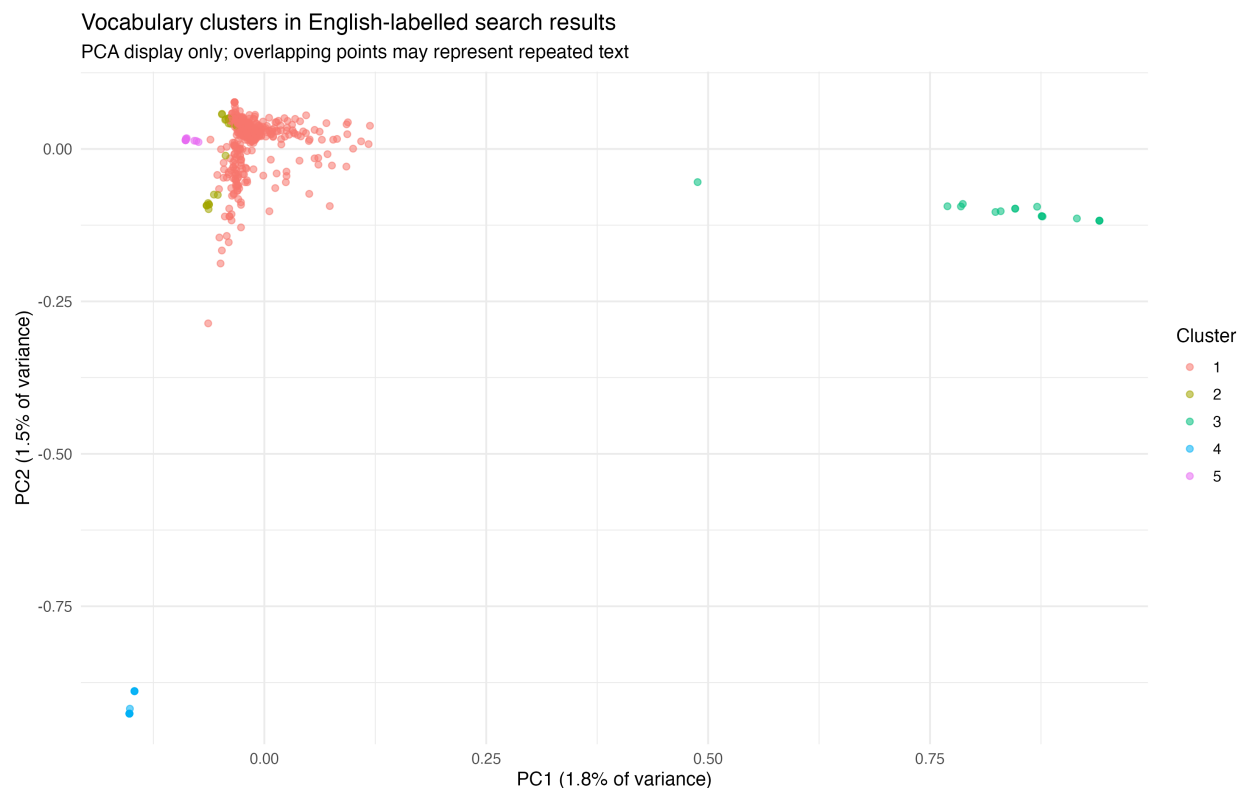


Figure 6: PCA display of clusters fitted in the full normalised TF-IDF space. Overlapping points can hide multiple posts.

The first two principal components explain only **1.78%** and **1.52%** of variance, respectively (3.30% together). The plot is therefore a very limited projection; visible proximity or separation cannot establish the full clustering quality. Empty-post exclusions, stemming, vocabulary threshold, query-term removal, k and random initialisation all affect the solution. We have not established stability across datasets or a unique optimal solution.

Together, the text and clustering results show that query vocabulary and repeated content shape the apparent differences between groups. The analysis identifies some recurring message families but provides **weak evidence for broad, distinct workplace-discussion clusters**. A full relevance review, more balanced sampling and sensitivity to repeated text would be needed before drawing stronger conclusions about working arrangements.

2.5 Network Analysis

Research question: How do accounts interact through replies around our three selected posts?

We collected responses around three selected posts about working from home vs returning to the office to analyse how people interacted when discussing mental health, related costs, remote work and job competition.

Data and graph construction

We requested the selected posts and up to two levels of replies, so we used replies to the selected post and replies to those replies. We did not request earlier posts higher up in the thread. We combined the saved responses while removing repeated posts using URIs, leaving 66 posts.

There were 62 different accounts in the network, and we added a connection from the replying account to the account it replied to when both posts were in the collected data. We treated the network as directed because a reply goes from one account to another, and the direction shows who replied to whom.

Connections were excluded where the account was replying to itself. If connections between the same accounts were repeated in the same direction, we counted them only once, leaving 60 directed connections.

Content review and scope

Reviewing the posts led us to label 33 posts as workplace discussion, 26 as context, 6 as unclear and only one as off topic.

We used these labels to describe content instead of removing posts from the network.

All collected accounts and reply connections stayed in the network. The results show interactions around 3 selected posts and are not meant to represent preferences across Bluesky as a whole.

Network structure

Table 7: Structure of the observed reply network

Measure	Value
Retrieved posts	66
Accounts	62
Directed edges	60
Directed density	0.0159
Isolated accounts	0
Weak components	3
Largest component: accounts	32

The network density was 0.0159. This means about 1.59% of the possible directed connections between different accounts were present.

The network had three connected groups, called weak components, containing 32, 19 and 11 accounts. We identified these groups by checking which accounts were connected while ignoring the direction of the arrows. We did not observe any connections between the three groups in the collected data.

Reply network across three selected conversations

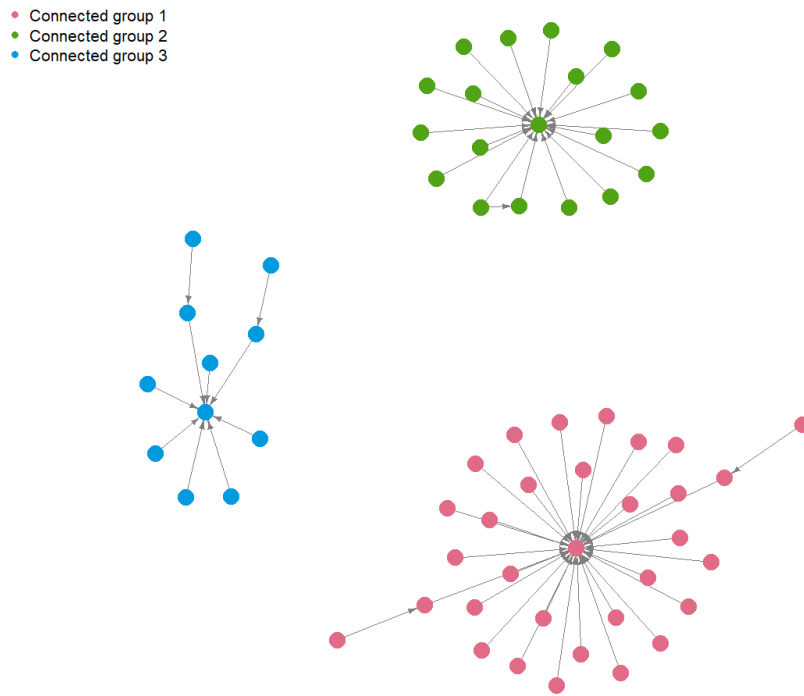


Figure 7: Observed reply network across three selected conversations (62 accounts, 60 edges). Arrows point from replying authors to parent authors; colours identify weak components.

Largest connected conversation

We chose the largest weak component to get a clearer view of the conversation with the most accounts. Its selected post discussed workplace isolation and mental health. This group contained 32 accounts and 31 directed connections. We calculated the network statistics and centrality scores using the full network.

Largest conversation: isolation and wellbeing

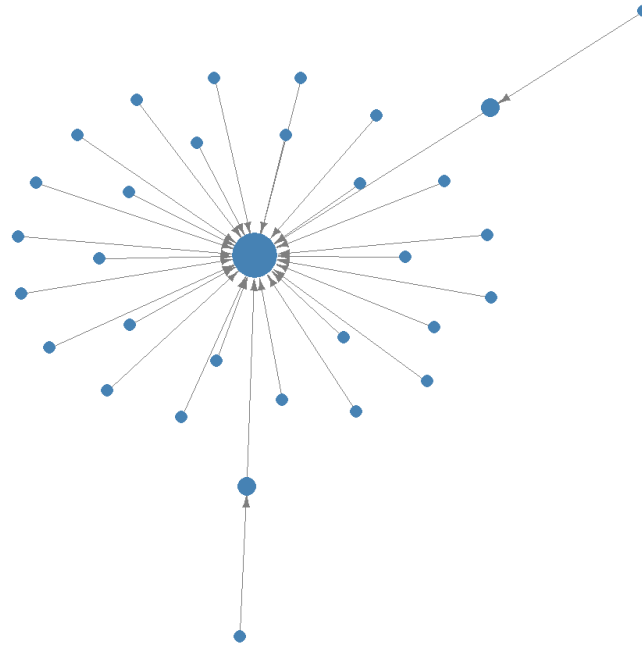


Figure 8: Largest weak component: isolation and wellbeing (32 accounts, 31 edges). Node size increases with in-degree.

Account centrality

We used two centrality measures: in-degree and raw, or unnormalised, directed betweenness. The table shows the three accounts with the highest in-degree and all accounts with a betweenness score above zero.

Table 8: Accounts with the three highest in-degrees or positive directed betweenness

Account	In-degree	Raw betweenness
cnn.com	29	0
rincewind.run	18	0
shockemc.bsky.social	8	0
goodboy.bsky.social	1	1
nasboat.org	1	1
nk-dizzle.bsky.social	1	1
trekkiebill.bsky.social	1	1

In-degree counts how many different accounts have a reply connection pointing to an author. The accounts cnn.com, rincewind.run and shockemc.bsky.social received connections from 29, 18 and eight different accounts respectively. These accounts wrote the three selected posts, so the way we collected the conversations favours them as reply targets. Replies indicate engagement within these conversations, but they do not establish agreement or endorsement.

Directed betweenness measures an account’s role on the shortest paths between other accounts, following the direction of the arrows. The accounts at the start and end of a path are not counted as intermediaries. The three accounts with the highest in-degree all had a betweenness score of zero. Receiving many replies therefore did not mean they acted as links between other accounts.

The accounts goodboy.bsky.social, nasboat.org, nk-dizzle.bsky.social and trekkiebill.bsky.social each had a raw betweenness score of one. These small scores show a limited role as intermediaries within the recorded reply paths. They do not show that these accounts connect the three separate groups or have influence across Bluesky.

Limitations

We used 3 selected posts to collect replies, so the network does not show all the reply connections these accounts may have. Our requests went as far as two reply levels, excluding posts from higher up in the thread. Before we added a reply connection, both posts had to be collected. Connections might therefore have been left out because the posts were outside our requests or because the API didn’t return them. The inclusion of more complete data could potentially change the groups and centrality scores.

Most replies were around the selected posts. With only two reply levels, there are few longer paths for betweenness to measure. This means the scores partly reflect how we collected the data. The 3 groups could be linked by replies we did not collect. They do not tell us which work arrangement people support or how common those opinions are across Bluesky.

The text and context did not give us enough information to label 6 posts more clearly. We did not check media, so that information is missing from our content review. We still kept all 66 posts, including the unclear and off-topic ones. An account’s colour or position on the graph does not tell us its workplace preference. These network review counts come from a separate sample from the group’s keyword-based content analysis.

3 Overall Findings

These sentiment percentages refer to the 548 scored posts (279 RTO and 269 WFH).

Overall, posts from the return-to-office search were more negative than posts from the remote-work search. 52.0% of the RTO posts were negative and 36.6% were positive, while for the WFH posts 26.8% were negative and 55.4% were positive. The chi-squared test showed this difference was unlikely to be due to chance. However, our text analysis and clustering suggest this doesn't mean people were just angrier about RTO. The two searches pulled in different kinds of posts, and a lot of the words that decided the sentiment scores came from repeated news stories and appeals, not from people giving their opinion on a work arrangement.

The text results showed policy and mandate words in the RTO search group, while the remote work group showed job and jobs. The clustering did not reveal clear workplace themes, as most posts were in one mixed cluster and the smaller groups held unrelated uses of search terms and repeated messages. These results suggest that some patterns came from how posts were collected and repeated. They do not establish whether people preferred remote work or RTO.

Our collection revealed three distinct conversational groups swirling around the selected posts, completely isolated from one another in the replies we collected. While the presence of replies signaled active engagement, it offered no window into sentiment or agreement. Because this dataset is strictly anchored to three specific posts, these snapshots reflect isolated moments of conversation rather than a sweeping portrait of views across Bluesky.

4 Conclusion

In this project we looked at Bluesky posts about return-to-office and remote work. Our answers to the four research questions are below.

1. **Sentiment.** Among scored posts, RTO posts were more often negative than WFH posts (52.0% compared with 26.8%). The chi-squared test found a significant association between search group and sentiment ($X^2 = 36.39$, $df = 2$, $p < .001$), with a small to moderate effect (Cramer's $V = 0.26$). The difference was clearly bigger (Cramer's $V = 0.46$) before we removed the search words from the dictionary, so this choice mattered.
2. **Vocabulary.** The RTO posts used more words like "policy", "mandates" and "employees", while the WFH posts used "job" and "jobs" more.
3. **Clusters.** We didn't find strong evidence of separate themes. Most posts (642 of 700, or 91.7%) ended up in one mixed cluster. The four small clusters were mostly repeated messages, and each came from just one search group.
4. **Network.** The network had 62 accounts and 60 connections, and a density of 0.016, so it was very sparse. Most replies went to the three accounts that wrote the selected posts, and the three conversations stayed separate from each other.

These results only describe the posts our searches returned. They don't tell us what people on Bluesky, or workers in general, actually prefer.

5 References

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Project repository: <https://github.com/AlexanderManalo19385596/comp3020-bluesky-rto-wfh>

Appendix: text and clustering code

01_data_collection.R

```
library(atrrrr)

# Run this once per session if not already authenticated:
# auth("your-handle.bsky.social")

posts_rto <- search_post("return to office", limit = 500, sort = "latest")
posts_wfh <- search_post("remote work", limit = 500, sort = "latest")

nrow(posts_rto)
nrow(posts_wfh)

saveRDS(posts_rto, "data/posts_rto_raw.rds")
saveRDS(posts_wfh, "data/posts_wfh_raw.rds")
```

02_data_cleaning.R

```
library(dplyr)
library(stringr)

# Load raw data collected in 01_data_collection.R
posts_rto <- readRDS("data/posts_rto_raw.rds") |> mutate(group = "RTO")
posts_wfh <- readRDS("data/posts_wfh_raw.rds") |> mutate(group = "WFH")

posts <- bind_rows(posts_rto, posts_wfh)

# --- Remove unsuitable records ---
posts_clean <- posts |>
  filter(!is.na(text)) |> # remove missing text
```

```

filter(str_length(text) > 10) |>      # remove near-empty posts (e.g. "no!!!!")
distinct(uri, .keep_all = TRUE)      # remove accidental duplicates

# Document how much was removed
nrow(posts)      # before cleaning
nrow(posts_clean) # after cleaning

# Save the cleaned combined dataset for everyone to use downstream
saveRDS(posts_clean, "data/posts_clean.rds")

```

08_sentiment_analysis.R

```

# 08_sentiment_analysis.R
#
# Research question: How do sentiment categories compare between the return-to-office
# and remote-work search groups? (Tested as: is sentiment associated with the search
# ↪ group?)
#
# H0: Search group (RTO/WFH) and sentiment (negative/neutral/positive) are independent.
# H1: Search group and sentiment are not independent (there is an association).
#
# Test: chi-squared test of independence (two categorical variables).
#
# Uses the same 727 English-labelled posts as 05_text_analysis.R and 07_clustering.R.
# Run 05_text_analysis.R first: it creates outputs/text/analysis_input.rds.
#
# Main analysis: the query words (return, office, remote, work, home, rto, wfh,
# remotework) and "trump" are removed from the sentiment dictionary before scoring.
# The Bing dictionary counts "work" and "trump" as positive. Many WFH posts contain
# "remote work", so leaving "work" in could make WFH posts look
# positive before anyone says anything.
#
# Robustness checks: (1) without removing those words, (2) all 1,137 cleaned posts
# (the original 03_hypothesis_testing.R analysis), (3) with job adverts removed.
#
# Outputs are saved to outputs/sentiment/. Object names start with sent_ so that
# sourcing this file inside the report does not overwrite other sections' objects.

library(dplyr)
library(tidyr)
library(stringr)
library(tidytext)
library(ggplot2)

dir.create("outputs/sentiment", recursive = TRUE, showWarnings = FALSE)

# -----
# Data: the English-labelled posts from the text analysis
# -----

sent_input <- readRDS("outputs/text/analysis_input.rds")

```

```

# clean_text has URLs and @mentions removed (done in 05_text_analysis.R)
sent_english <- sent_input$posts |>
  transmute(uri, group, text = clean_text)

if (nrow(sent_english) != 727) {
  warning("Expected 727 English-labelled posts but found ", nrow(sent_english),
    ". Check that 05_text_analysis.R is the final version.")
}

# Words removed from the dictionary in the main analysis
sent_drop_words <- c(sent_input$topic_words, "trump")

posts_clean <- readRDS("data/posts_clean.rds")

# Post dates (Bluesky's indexed time) so we can report time coverage by group
date_col <- intersect(c("indexed_at", "created_at"), names(posts_clean))[1]
if (is.na(date_col)) stop("No date column found in data/posts_clean.rds")

sent_dates <- data.frame(
  uri = posts_clean$uri,
  post_date = as.Date(substr(as.character(posts_clean[[date_col]]), 1, 10))
)

sent_date_range <- sent_english |>
  left_join(sent_dates, by = "uri") |>
  group_by(group) |>
  summarise(
    first_post = min(post_date, na.rm = TRUE),
    last_post = max(post_date, na.rm = TRUE),
    .groups = "drop"
  )

# -----
# Sentiment scoring
# -----
# Each word is matched to the Bing lexicon (positive or negative). A post is
# Positive if it has more positive than negative words, Negative if more negative
# than positive, and Neutral if the counts are equal. Posts with no lexicon word
# cannot be scored and are left out of the test.

sent_score <- function(df, drop_words = character(0)) {
  lexicon <- get_sentiments("bing") |> filter(!word %in% drop_words)
  df |>
    select(uri, group, text) |>
    unnest_tokens(word, text) |>
    inner_join(lexicon, by = "word") |>
    count(uri, group, sentiment) |>
    pivot_wider(names_from = sentiment, values_from = n, values_fill = 0) |>
    mutate(overall_sentiment = case_when(
      positive > negative ~ "Positive",
      negative > positive ~ "Negative",
      TRUE ~ "Neutral"
    ))
}

```

```

# Which of the removed words does the Bing dictionary actually score, and how often?
sent_drop_counts <- sent_english |>
  unnest_tokens(word, text) |>
  filter(word %in% sent_drop_words) |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(group, word, sentiment, name = "occurrences")

# -----
# Main analysis: 727 English-labelled posts, query words and "trump" removed
# -----

sent_scored <- sent_score(sent_english, sent_drop_words)

# Contingency table: search group (rows) by sentiment (columns)
sent_X <- table(sent_scored$group, sent_scored$overall_sentiment)

sent_test <- chisq.test(sent_X)

# Simulated p-value, as shown in the lecture slides
set.seed(3020)
sent_sim <- chisq.test(sent_X, simulate.p.value = TRUE, B = 10000)

# Effect size: Cramer's V (0 = no association, 1 = perfect association)
sent_V <- sqrt(unnamed(sent_test$statistic) / (sum(sent_X) * (min(dim(sent_X)) - 1)))

sent_pct_table <- round(100 * prop.table(sent_X, 1), 1) # row percentages

sent_coverage <- sent_english |>
  count(group, name = "english_posts") |>
  left_join(count(sent_scored, group, name = "scored_posts"), by = "group") |>
  mutate(
    unscored = english_posts - scored_posts,
    percent_scored = round(100 * scored_posts / english_posts, 1)
  )

# Which lexicon words drive each sentiment class?
sent_top_words <- sent_english |>
  unnest_tokens(word, text) |>
  inner_join(get_sentiments("bing") |> filter(!word %in% sent_drop_words), by = "word")
  ↪ |>
  count(group, sentiment, word, sort = TRUE) |>
  group_by(group, sentiment) |>
  slice_head(n = 5) |>
  summarise(top_words = paste0(word, " (", n, ")", collapse = ", "), .groups = "drop")

# -----
# Robustness check 1: same posts, but without removing the query words / "trump"
# -----

sent_scored_raw <- sent_score(sent_english)
sent_X_raw <- table(sent_scored_raw$group, sent_scored_raw$overall_sentiment)
sent_test_raw <- chisq.test(sent_X_raw)
sent_pct_raw <- round(100 * prop.table(sent_X_raw, 1), 1)

```

```

# -----
# Robustness check 2: all 1,137 cleaned posts (original 03 analysis, no removal)
# -----

sent_scored_all <- sent_score(posts_clean)
sent_X_all <- table(sent_scored_all$group, sent_scored_all$overall_sentiment)
sent_test_all <- chisq.test(sent_X_all)
sent_pct_all <- round(100 * prop.table(sent_X_all, 1), 1)

# Should match the earlier result: RTO 203/54/171, WFH 36/73/412
if (!all(as.vector(sent_X_all) == c(203, 36, 54, 73, 171, 412))) {
  warning("All-posts table differs from the earlier 03 result (203/54/171 and
    ↪ 36/73/412).")
}

# -----
# Robustness check 3: remove job adverts (main scoring rule otherwise unchanged)
# -----
# Job adverts use positive words and are not opinions about work arrangements.
# This is a keyword rule, not a manual label, so it is only a check.

sent_job_pattern <- paste0(
  "\\b(hiring|apply now|apply here|apply today|vacanc(y|ies)|recruit(ing|er|ment)?|",
  "careers?|remotejobs|job
  ↪ (opening|openings|alert|alerts|posting|board|listing|listings)|salary)\\b"
)

sent_english$job_ad_like <- str_detect(str_to_lower(sent_english$text), sent_job_pattern)

# Note: the new column is called job_ads so the percentage uses the logical column
sent_job_share <- sent_english |>
  group_by(group) |>
  summarise(
    posts = n(),
    job_ads = sum(job_ad_like),
    percent = round(100 * mean(job_ad_like), 1),
    .groups = "drop"
  )

sent_scored_nojobs <- sent_score(filter(sent_english, !job_ad_like), sent_drop_words)
sent_X_nojobs <- table(sent_scored_nojobs$group, sent_scored_nojobs$overall_sentiment)
sent_test_nojobs <- chisq.test(sent_X_nojobs)
sent_pct_nojobs <- round(100 * prop.table(sent_X_nojobs, 1), 1)

# -----
# Figure and saved tables
# -----

sent_plot_df <- as.data.frame(sent_pct_table, stringsAsFactors = FALSE)
names(sent_plot_df) <- c("group", "sentiment", "percent")
sent_plot_df$sentiment <- factor(sent_plot_df$sentiment,
  levels = c("Negative", "Neutral", "Positive"))

```

```

sent_plot <- ggplot(sent_plot_df, aes(x = group, y = percent, fill = sentiment)) +
  geom_col(width = 0.6) +
  geom_text(
    aes(label = paste0(percent, "%"), colour = sentiment),
    position = position_stack(vjust = 0.5), size = 3.5
  ) +
  scale_fill_manual(values = c(Negative = "#C2410C", Neutral = "#8A94A0", Positive =
    ↪ "#0A8F83")) +
  scale_colour_manual(values = c(Negative = "white", Neutral = "black", Positive =
    ↪ "white"),
                      guide = "none") +
  labs(
    title = "Sentiment of English-labelled search results",
    subtitle = "Share of scored posts; query words and \"trump\" not scored",
    x = NULL, y = "% of scored posts", fill = NULL
  ) +
  theme_minimal(base_size = 12)

ggsave("outputs/sentiment/sentiment_by_group.png", sent_plot,
       width = 6.5, height = 4.5, dpi = 300, bg = "white")

write.csv(as.data.frame.matrix(sent_X), "outputs/sentiment/contingency_table.csv")
write.csv(as.data.frame.matrix(round(sent_test$stdres, 2)),
  ↪ "outputs/sentiment/standardised_residuals.csv")
write.csv(sent_coverage, "outputs/sentiment/coverage.csv", row.names = FALSE)
write.csv(sent_job_share, "outputs/sentiment/job_ad_share.csv", row.names = FALSE)
write.csv(sent_top_words, "outputs/sentiment/top_lexicon_words.csv", row.names = FALSE)
write.csv(sent_drop_counts, "outputs/sentiment/removed_word_counts.csv", row.names =
  ↪ FALSE)
write.csv(as.data.frame.matrix(sent_X_raw),
  ↪ "outputs/sentiment/contingency_table_unadjusted.csv")
writeLines(capture.output(sessionInfo()), "outputs/sentiment/session_info.txt")

# -----
# Small helpers used by the report text
# -----

sent_pct <- function(x) paste0(format(round(x, 1), nsmall = 1), "%")

sent_p <- function(p) {
  if (p < 0.001) "p < .001" else paste0("p = ", format(round(p, 3), nsmall = 3))
}

# Print the key results when run on its own
print(sent_drop_counts)
print(sent_coverage)
print(sent_X)
print(sent_pct_table)
print(sent_test)
print(sent_sim)
print(round(sent_test$stdres, 2))
print(sent_V)
print(sent_top_words)
print(sent_X_raw)

```

```

print(sent_pct_raw)
print(sent_test_raw)
print(sent_X_all)
print(sent_test_all)
print(sent_job_share)
print(sent_X_nojobs)
print(sent_pct_nojobs)
print(sent_test_nojobs)

```

05_text_analysis.R

```

# RQ: How does vocabulary differ between English-labelled RTO/WFH search results?
library(dplyr)
library(tidyr)
library(tidytext)
library(stringr)
library(ggplot2)
dir.create("outputs/text", recursive = TRUE, showWarnings = FALSE)

# Language metadata is supplied by Bluesky; missing/mixed tags are not English-only.
posts <- readRDS("data/posts_clean.rds")
language_codes <- lapply(posts$langs, function(x) tolower(unlist(x)))
audit <- posts |> select(uri, group, text)
audit$language <- vapply(language_codes, paste, collapse = ";", FUN.VALUE = "")
audit$english <- vapply(language_codes, function(x) {
  length(x) > 0 && all(grepl("^en(-[a-z]+)?$", x))
}, logical(1))
# Remove URLs (also bare domains/short links) and account mentions before tokenising.
url_pattern <-
  ↪ regex("https?://\\S+|www\\.\\S+|\\b[a-z0-9][a-z0-9.-]*\\. [a-z]{2,}(:[/?#]\\S*)?",
  ↪ TRUE)
audit$clean_text <- audit$text |>
  str_remove_all(url_pattern) |>
  str_remove_all("@[[:alnum:]]_\\.+") |>
  str_replace_all("[']", "")
words <- audit |> filter(english) |> select(uri, group, clean_text) |>
  unnest_tokens(word, clean_text) |>
  anti_join(stop_words, by = "word") |>
  filter(str_detect(word, "[a-z]+$"), nchar(word) >= 2)
audit <- audit |> mutate(
  usable = english & uri %in% words$uri,
  decision = case_when(!english & language == "" ~ "Missing language tag",
    !english ~ "Non-English or mixed language tags",
    !usable ~ "No retained English word tokens", TRUE ~ "Included")
)
text_posts <- audit |> filter(usable)
text_counts <- audit |> count(group, decision, name = "posts")
word_counts <- words |> count(group, word, sort = TRUE)
# These are search-result groups, not verified opinions or a relevance-screened corpus.
topic_words <- c("return", "office", "remote", "work", "home", "rto", "wfh",
  ↪ "remotework")
post_word_rates <- words |> filter(!word %in% topic_words) |>
  distinct(uri, group, word) |> count(group, word, name = "posts_with_word") |>

```

```

left_join(text_posts |> count(group, name = "total_posts"), by = "group") |>
mutate(percent = 100 * posts_with_word / total_posts)

# Both figures share a numeric scale across groups; top 15 terms are chosen per group.
plot_words <- function(data, metric, title, axis_label) {
  top <- data |> mutate(value = .data[[metric]]) |> group_by(group) |>
  arrange(desc(value), word, .by_group = TRUE) |> slice_head(n = 15) |> ungroup()
  ggplot(top, aes(reorder_within(word, value, group), value, fill = group)) +
  geom_col(show.legend = FALSE) + facet_wrap(~group, scales = "free_y") +
  scale_x_reordered() + coord_flip() + theme_minimal(base_size = 12) +
  scale_fill_manual(values = c(RTO = "#CC6677", WFH = "#228899")) +
  labs(title = title, subtitle = "English-labelled search results; relevance not fully
  ↪ screened",
  x = NULL, y = axis_label)
}
word_plot <- plot_words(word_counts, "n", "Frequent words in RTO and WFH search results",
  ↪ "Word occurrences")
rate_plot <- plot_words(post_word_rates, "percent", "Vocabulary beyond the topic terms",
  ↪ "Percentage of usable posts")
print(word_plot)
print(rate_plot)
ggsave("outputs/text/word_frequency.png", word_plot, width = 11, height = 7, dpi = 300,
  ↪ bg = "white")
ggsave("outputs/text/post_word_percentages.png", rate_plot, width = 11, height = 7, dpi =
  ↪ 300, bg = "white")
write.csv(audit, "outputs/text/inclusion_audit.csv", row.names = FALSE)
write.csv(text_counts, "outputs/text/sample_counts.csv", row.names = FALSE)
write.csv(word_counts, "outputs/text/word_counts.csv", row.names = FALSE)
write.csv(post_word_rates, "outputs/text/post_word_percentages.csv", row.names = FALSE)
saveRDS(list(posts = text_posts, words = words, topic_words = topic_words),
  ↪ "outputs/text/analysis_input.rds")
writeLines(capture.output(sessionInfo()), "outputs/text/session_info.txt")
print(text_counts)

```

07_clustering.R

```

# RQ: Do English-labelled search results contain distinct vocabulary clusters?
# Run 05_text_analysis.R first. Module 6: TF-IDF, k-means and elbow analysis.
library(dplyr)
library(tidytext)
library(SnowballC)
library(ggplot2)
dir.create("outputs/clustering", recursive = TRUE, showWarnings = FALSE)
a <- readRDS("outputs/text/analysis_input.rds")
# Remove query-defining words and stem English words; keep terms in >=3 posts.
counts <- a$words |> filter(!word %in% a$topic_words) |>
  mutate(term = wordStem(word, language = "english")) |> count(uri, term, name = "n")
vocabulary <- counts |> count(term, name = "documents") |> filter(documents >= 3)
counts <- counts |> semi_join(vocabulary, by = "term")
M <- unclass(xtabs(n ~ uri + term, counts))
# TF = within-post term proportion; IDF = log(N / document frequency).
T <- sweep(M / rowSums(M), 2, log(nrow(M) / colSums(M > 0)), "*")

```

```

stopifnot(all(rowSums(T^2) > 0), all(is.finite(T)))
X <- T / sqrt(rowSums(T^2))
# Ordinary k-means uses squared Euclidean distance on unit-length TF-IDF rows.
# Pairwise squared Euclidean distance / 2 equals cosine dissimilarity here.
# This is NOT spherical k-means: centroids are not constrained to unit length.
fits <- lapply(1:10, function(k) {
  set.seed(3020 + k)
  kmeans(X, centers = k, nstart = 30, iter.max = 100)
})
elbow <- data.frame(k = 1:10, within_ss = sapply(fits, function(f) f$tot.withinss))
elbow$reduction <- c(NA, -diff(elbow$within_ss))
# No decisive elbow: k=5 separates repeated referral/news families mixed at k=3.
# This is an interpretable exploratory cut, not evidence of five natural themes.
chosen_k <- 5
candidate_summary <- bind_rows(lapply(2:10, function(k) {
  bind_rows(lapply(1:k, function(j) {
    idx <- which(fits[[k]]$cluster == j)
    groups <- a$posts$group[match(rownames(X)[idx], a$posts$uri)]
    terms <- names(sort(colMeans(T[idx, , drop = FALSE]), decreasing = TRUE))[1:8]
    data.frame(k = k, cluster = j, posts = length(idx), RTO = sum(groups == "RTO"),
              WFH = sum(groups == "WFH"), top_stems = paste(terms, collapse = ", "))
  })))
}))
fit <- fits[[chosen_k]]
# Relabel in decreasing size so C1 is always the largest cluster.
cluster <- match(fit$cluster, order(fit$size, decreasing = TRUE))
assignments <- a$posts[match(rownames(X), a$posts$uri), c("uri", "group", "text")]
assignments$cluster <- cluster
centres <- do.call(rbind, lapply(1:chosen_k, function(j) colMeans(X[cluster == j, , drop
  ↪ = FALSE])))
assignments$distance_to_centre <- rowSums((X - centres[cluster, , ])^2)
top_terms <- bind_rows(lapply(1:chosen_k, function(j) {
  data.frame(cluster = j, term = colnames(T), weight = colMeans(T[cluster == j, , drop =
  ↪ FALSE])) |>
  arrange(desc(weight), term) |> slice_head(n = 10)
}))
examples <- assignments |> group_by(cluster) |>
  arrange(distance_to_centre, uri, .by_group = TRUE) |> slice_head(n = 5) |> ungroup()
cluster_summary <- assignments |> count(cluster, group) |> group_by(cluster) |>
  mutate(cluster_size = sum(n), percent = 100 * n / cluster_size) |> ungroup()
# Diagnose identical processed text within clusters; different URIs remain separate
↪ posts.
repetition <- a$words |> group_by(uri) |> summarise(signature = paste(word, collapse = "
  ↪ ")) |>
  inner_join(assignments |> select(uri, cluster), by = "uri") |> group_by(cluster) |>
  summarise(posts = n(), unique_token_sequences = n_distinct(signature), .groups =
  ↪ "drop")
# PCA is for display only. Clusters were fitted in all retained term dimensions.
pca <- prcomp(X, center = TRUE, scale. = FALSE, rank. = 2)
variance <- 100 * pca$sdev^2 / sum(pca$sdev^2)
coords <- data.frame(PC1 = pca$x[, 1], PC2 = pca$x[, 2], cluster = factor(cluster))
elbow_plot <- ggplot(elbow, aes(k, within_ss)) + geom_line() + geom_point() +
  geom_vline(xintercept = chosen_k, linetype = "dashed", colour = "#CC6677") +
  scale_x_continuous(breaks = 1:10) + theme_minimal(base_size = 12) +

```

```

labs(title = "Cluster-count comparison", subtitle = "No decisive elbow; k = 5 used as
↳ an exploratory summary",
      x = "Number of clusters", y = "Within-cluster sum of squares")
cluster_plot <- ggplot(coords, aes(PC1, PC2, colour = cluster)) +
  geom_point(alpha = 0.55, size = 1.7) + theme_minimal(base_size = 12) +
  labs(title = "Vocabulary clusters in English-labelled search results",
        subtitle = "PCA display only; overlapping points may represent repeated text",
        x = sprintf("PC1 (%.1f%% of variance)", variance[1]),
        y = sprintf("PC2 (%.1f%% of variance)", variance[2]), colour = "Cluster")
term_plot <- ggplot(top_terms, aes(reorder_within(term, weight, cluster), weight)) +
  geom_col(fill = "#228899") + facet_wrap(~cluster, scales = "free_y") +
  scale_x_reordered() + coord_flip() + theme_minimal(base_size = 12) +
  labs(title = "Terms characterising each cluster", x = NULL, y = "Mean TF-IDF weight",
        subtitle = "Stemmed terms; mean weights calculated within each cluster")
for (name in c("elbow", "cluster", "term")) {
  figure <- get(paste0(name, "_plot"))
  print(figure)
  ggsave(paste0("outputs/clustering/", name, ".png"), figure, width = 11, height = 7, dpi
  ↳ = 300, bg = "white")
}
excluded <- a$posts |> filter(!uri %in% assignments$uri) |>
  select(uri, group, text) |> mutate(reason = "No terms after topic-word and
  ↳ document-frequency filters")
metrics <- data.frame(posts = nrow(X), terms = ncol(X), k = chosen_k,
  excluded_after_text = nrow(excluded), explained_between_percent = 100 * fit$betweenss
  ↳ / fit$totss,
  pc1_percent = variance[1], pc2_percent = variance[2])
# Save all result tables using the same output convention.
results <- list(assignments = assignments, top_terms = top_terms, examples = examples,
  cluster_summary = cluster_summary, elbow = elbow, repetition = repetition,
  candidate_summary = candidate_summary, excluded_posts = excluded, metrics = metrics)
for (name in names(results)) write.csv(results[[name]],
  paste0("outputs/clustering/", name, ".csv"), row.names = FALSE)
saveRDS(list(fit = fit, cluster = cluster, uris = rownames(X), terms = colnames(X)),
  "outputs/clustering/model.rds")
writeLines(capture.output(sessionInfo()), "outputs/clustering/session_info.txt")
print(metrics)
print(cluster_summary)

```